

Design of IOT Architecture for Battery Charging Management System

A DISSERTATION

*submitted in partial fulfillment of the
requirements for the award of the degree of*

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Submitted by

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CERTIFICATE

I hereby certify that the work which is being presented in the M.Tech. Dissertation entitled “**Design of IOT Architecture for Battery Charging Management System**”, in partial fulfillment of the requirements for the award of the **Master of Technology in Internet of Things, Department of Electronics and Communication Engineering** is an authentic record of my own work carried out during a period from 05-08-2024 to 31-04-2025 under the supervision of **Dr. Sujith Kalluri Ph.D, Associate Professor**, Department of Electronics and Communication Engineering.

The matter presented in this thesis has not been submitted for the award of any other degree elsewhere.

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Index

Content	Page.no
Certificate.....	i
Acknowledgments.....	ii
Table of Contents.....	iii - v
List of Figures.....	vi
List of Software & Tools.....	vii
List of Simulations.....	vii
Data Monitoring & Visualization.....	viii
Abstract.....	.ix
Abbreviations.....	x

Chapter – 1

Introduction.....	1 – 7
1.1 Background and Motivation.....	4
1.2 Current State of EV Charging Technology.....	4
1.3 Challenges in Renewable Energy Integration.....	5
1.4 Role of IoT in Smart Charging Systems.....	5
1.5 Research Objectives and Contributions.....	6
1.6 Our Contribution.....	6
1.7 Thesis Organization.....	7

Chapter – 2

Power Generation Integration and Automation with PCC.....	8 – 13
2.1 Solar Energy.....	8
2.2 Energy Storage Systems (ESS).....	10
2.3 Grid Power.....	11
2.4 Point of Common Coupling (PCC).....	13

Chapter – 3

LITERATURE REVIEW	14 – 15
3.1 Solar Monitoring and Energy Harvesting.....	14
3.2 IoT in EV Charging Infrastructure.....	15
3.3 Energy Storage Systems (ESS) and Renewable Integration.....	15
3.4 Power Quality and Grid Integration.....	16
3.5 Fault Detection and State-of-Charge Optimization.....	16
3.6 Battery Swapping and Unit Charging.....	17

Chapter – 4

Problem Statement and Proposed System Methodology	18 – 27
4.1 Problem Statement in EV Charging Infrastructure.....	18
4.2 Proposed System Methodology.....	19
4.2.1 proposed IOT Architecture layers.....	20
4.2.2. End to End power flow.....	22
4.2.3. PCC Integration and Automation.....	23
4.2.4 Algorithms.....	24
• Solar MPPT	
• Battery Swapping Charging	
• Vehicle Charging control calculation	

Chapter – 5

Software Simulations and Tools.....	28 – 30
5.1 MATLAB.....	28
5.2 NODE- RED.....	29
5.3 AWS – Amazon Web Services.....	30
• MQTT Protocol Integration	
• Data Storage with Amazon S3	
• Email Alerts	
• Data Visualization with Amazon QuickSight	

Chapter – 6

Design Simulation and Results.....	33 – 49
6.1 Design Simulation Battery charging Technology and Power Integration...	33
6.2 Solar Generation Unit.....	34
6.3 Grid Generation Unit.....	35
6.4 ESS Energy Storage system.....	36
6.5 PCC – Point of common coupling and Power Source Integration and Automation.....	38
6.6 Battery Swapping System.....	39
6.7 Dc Fast Charging.....	41
6.8 Simulation Monitoring Data.....	42
6.9 Node Red.....	43
6.10.1 Solar Visualization.....	44
6.10.2 ESS Visualization.....	45
6.10.3 Grid Power Visualization.....	46
6.10.4 PCC Visualization.....	46
6.10.5 Battery Swapping and Fast charging Station Visualization.....	47
6.11 AWS – Integration.....	47
6.11.1 Data Storage in Amazon S3.....	48
6.11.2 Email Alerts with Amazon SNS.....	48
6.11.3 Visualization with Amazon QuickSight.....	49
Estimated Calculated Spread sheet.....	50 - 51

Chapter – 7

Conclusion.....	52
Future Scope.....	52

Chapter – 8

References.....	53- 54
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Chapter – 9

Publication.....	55
------------------	----

List of Figures

Fig 1 : EV Charging Drivers and Challenges.....	1
Fig 2 : Smart charging.....	2
Fig 3 : Charging infra IOT integration.....	3
Fig 4 : Power sources integration.....	8
Fig 5 : Solar EV Charging.....	9
Fig 6 : Solar with MPPT Unit.....	10
Fig 7 : Battery Energy Storage System.....	11
Fig 8 : Grid EV Charging.....	12
Fig 9 : Designed 5 Layered Architecture.....	18
Fig 10 : PCC Control Flow.....	21
Fig 11 : PPC integration and Automation.....	21

List of Software and Tools

Fig 12 : MATLAB Tool.....	26
Fig 13 : Node-Red Tool.....	27
Fig 14 : AWS Tool.....	28

List of Simulations

Fig 15 : Simulation of Battery charging Technology and Power Integration...	31
Fig 16 : Solar generation unit.....	32
Fig 17 : Grid Generation unit.....	33
Fig 18 : BESS Battery Energy Storage System.....	35
Fig 19 : PCC Point of Coming Coupling.....	36
Fig 20 : Battery Swapping Station.....	38
Fig 21 : DC Fast Charging Station.....	39
Fig 22 : Parameters Data monitoring.....	40

Data monitoring and Visualization

Fig 23 : Node-Red Data mapping and monitoring.....	41
Fig 24 : Solar Data Visualization.....	43
Fig 25 : ESS Visualization.....	43
Fig 26 : Grid Power Visualization.....	44
Fig 27 : PCC Visualization.....	44
Fig 28 : Battery Swapping Station Visualization.....	45
Fig 29 : DC Fast Charging Station Visualization.....	45
Fig 30 : Data storage in AWS S3 Bucket.....	46
Fig 31 : AWS SNS Email Alerts.....	46
Fig 32: AWS QuickSights flow.....	47

ABSTRACT

This research proposes a novel IoT-based architecture for electric vehicle (EV) battery charging management, addressing the critical demand for scalable and sustainable charging infrastructure. The system integrates solar energy, energy storage systems (ESS), and grid power through a Point of Common Coupling (PCC), optimizing energy distribution via a five-layer IoT architecture: Sensor, Microcontroller & Data Acquisition, Communication, Cloud Server, and User Interface layers. Real-time monitoring, predictive analytics, and automated decision-making enhance operational efficiency, achieving 88.6% energy-saving efficiency (1,608 kW/day from renewables) and 88.5% cost-saving efficiency (₹19,296/day savings). The architecture supports battery swapping stations and DC fast charging, generating ₹26,986/day in revenue while stabilizing grid loads. This work bridges gaps in EV infrastructure standardization and economic viability, offering a scalable solution for smart cities and industrial applications.

Keywords: IoT Architecture, Renewable Energy Integration, Battery Swapping, Smart Grid, Predictive Analytics, Energy Storage Systems, Real-time Monitoring.

Abbreviations

EVs : Electric Vehicles

ESS : Energy Storage Systems

IoT : Internet of Things

PCC : Point of Common Coupling

MPPT : Maximum Power Point Tracking

SOC : State of Charge

CI : Charging Infrastructure

RES : Renewable Energy Sources

BSS : Battery Swapping Stations

GP : Grid Power

PA : Predictive Analytics

SCI : Smart City Initiatives

BESS : Battery Energy Storage System

RTDM : Real-Time Data Monitoring

OM : Operational Metrics

AWS : Amazon Web Services